

MATH 8670 Numerical methods for partial differential equations

Spring 2027

Time: TBA

Room: TBA

General Information

Instructor: Dr. Qingshan Chen
Office: Martin O-201
Office hours: 3-4pm, Friday
Contact information: qsc@clemson.edu or 656-4565

Overview

This course covers numerical methods for partial differential equations. Topics include classical ones such as algorithmic development and implementation, stability analysis, convergence analysis, and error analysis, as well as the solution of PDEs by modern machine learning techniques. The types of equations considered in this course, in the order of priority, include elliptic equations, hyperbolic equations, parabolic equations, and, time permitting, some nonlinear models from various application areas, such as the Navier Stokes equations, shallow water equations. *This course encourages interactive-style code developments, using Matlab Live Script or Python Jupyter Notebook, and existing code bases/templates.*

Recommended References

1. Strikwerda, *Finite difference schemes and partial differential equations* (Text-book for the class)
2. Issacson & Keller, *Analysis of Numerical Methods*
3. Evans, *Partial Differential Equations*

Learning outcomes

At the end of the semester, the student should be able to:

1. Understand crucial concepts in numerical analysis such as stability, consistency, and convergence;
2. Understand the properties of basic types of partial differential equations;
3. Compute the Fourier transform of common functions, and apply the technique to solve partial differential equations;
4. Derive, analyze, and implement finite difference schemes for 1D and 2D elliptic problems;

5. Derive, analyze, and implement finite difference schemes for elliptic eigenvalue problems;
6. Derive, analyze, and implement finite difference schemes for parabolic problems;
7. Derive, analyze, and implement finite difference schemes for hyperbolic problems.
8. Understand the basic concepts of machine learning techniques as applied to PDEs.

Covered Topics (Tentative)

- Preliminaries
- 1D & 2D Elliptic boundary value problems
- Sturm-Liouville eigenvalue problems
- Fourier transform and discrete Fourier transform
- Discretization of
 - parabolic equations
 - wave equations
 - geophysical flows
- Machine learning for PDEs and physical models

Learning/Teaching Mode

The class is conducted primarily through in-person lectures and office hours. However, the class may turn temporarily into a virtual mode when the circumstances warrant, such as inclement weather, etc. All changes will be announced in advance through emails.

Attendance Policy

As graduate students, you are expected to attend the class regularly. Use of cell phones and pagers is prohibited during the class meeting time.

Per CU policy, if the instructor or a guest lecturer is late, you are expected to wait 15 minutes before leaving.

Any exam that was scheduled at the time of a class cancellation due to inclement weather will be given at the next class meeting unless contacted by the instructor. Any assignments due at the time of a class cancellation due to inclement weather will be due at the next class meeting unless contacted by the instructor. Any extension or postponement of assignments or exams must be granted by the instructor via email or Blackboard within 24 hours of the weather related cancellation.

Course Assessment

Any copying of code or answers from other students, from work that students have done in previous semesters, from solution keys that have been generated by instructors at this or any other institution, or from the web is considered plagiarism, and will result in a zero score for the affected assignment, and a five-point reduction from your averaged final score.

There will be weekly or biweekly homework assignments, which involve both analytical problems, and computational exercises. Late homework or assignment is **not accepted**. Perfect grades on either homeworks or exams require **full and thorough** explanations of all of your answers. All code used to generate solutions must be included as part of the material that is submitted.

There will be one to two term projects, which involve analysis of the problems at hand, and adaptation and extension of given code bases for the problems.

Final grades will be determined based on the following percentages:

Assignment	Weight
Homework assignments:	60%
Term project(s):	40%

Grades will be assigned as follows:

Grading Scale	Letter Grade
90+	A
75-89	B
60-74	C
below 60	F

Accommodations for Students with Disabilities

Students with disabilities who need accommodations should make an appointment with Dr. Arlene Stewart, Director of Disability Services, to discuss specific needs within the first month of classes. Students should present an Academic Accommodation Letter from Student Disability Services when they meet with instructors. Student Disability Services is located in Suite 239 Academic Success Building (656-6848; sds-l@clemson.edu). Please be aware that accommodations are not retroactive and new Academic Accommodation Letters must be presented each semester.

Clemson University Title IX (Sexual Harassment) Statement

Clemson University is committed to a policy of equal opportunity for all persons and does not discriminate on the basis of race, color, religion, sex, sexual orientation, gender, pregnancy, national origin, age, disability, veteran's status, genetic information or protected activity (e.g., opposition to prohibited discrimination or participation in any complaint process, etc.) in employment, educational programs and activities,

admissions and financial aid. This includes a prohibition against sexual harassment and sexual violence as mandated by Title IX of the Education Amendments of 1972.

The University is committed to combating sexual harassment and sexual violence. As a result, you should know that University faculty and staff members who work directly with students are required to report any instances of sexual harassment and sexual violence, to the University's Title IX Coordinator. What this means is that as your professor, I am required to report any incidents of sexual harassment, sexual violence or misconduct, stalking, domestic and/or relationship violence that are directly reported to me, or of which I am somehow made aware.

There are two important exceptions to this requirement about which you should be aware:

- Confidential Resources and facilitators of sexual awareness programs such as "Take Back the Night and Aspire to be Well" when acting in those capacities, are not required to report incidents of sexual discrimination.
- Another important exception to the reporting requirement exists for academic work. Disclosures about sexual harassment, sexual violence, stalking, domestic and/or relationship violence that are shared as part of an academic project, a research project, classroom discussion, or course assignment, are not required to be disclosed to the University's Title IX Coordinator.

This policy is located at <http://www.clemson.edu/campus-life/campus-services/access/title-ix/>. Ms. Alesia Smith is the Executive Director for Equity Compliance and the Title IX Coordinator. Her office is located at 223 Holtzendorff Hall, phone number is 864.656.3181, and email address is alesias@clemson.edu.

Academic Integrity

As graduate students, you are expected to adhere to the official CU academic integrity policy, which is available at

<http://gradspace.editme.com/AcademicGrievancePolicyandProcedures#integritypolicy>

The following is the official statement on academic integrity at Clemson University.

As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning". Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form.